

Expectation, Variance, Skewness, and Kurtosis

Distribution moments, payoff shape, portfolio risk, instability, and point-in-time statistics.

Library of the Quant

Educational reference manual

Manual Profile

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Educational Use Only

This manual is educational material for quant research study. It is not financial advice, not a trading recommendation, and not a guarantee of correctness or live trading usefulness.

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How to use this manual

BEGINNER

1. DEFINE THE VARIABLE FIRST

State the outcome, unit, horizon, population, and information cutoff. A moment without a variable contract is a number without a stable interpretation.

2. READ MOMENTS AS A SEQUENCE

Expectation locates a probability-weighted center; variance measures squared spread; skewness adds directional asymmetry; kurtosis emphasizes extreme deviations and mixture shape.

3. CHECK EXISTENCE AND ESTIMATION

A sample calculation always returns a finite result, but the modeled population moment may not exist. Inspect tails, effective sample size, dependence, and regime stability.

4. CONNECT FORMULAS TO MECHANISMS

Use the market example and contribution view to identify which states or dates drive the statistic. Never stop at a scalar dashboard tile.

5. VALIDATE POINT IN TIME

Rolling windows, standardization, adjustments, and universe membership must respect the decision timestamp. Recompute every feature exactly as it would have existed live.

6. PREFER A PANEL, NOT ONE MOMENT

Pair mean and variance with quantiles, drawdowns, expected shortfall, sample size, uncertainty, and robust alternatives. Different shapes can share identical first four moments.

DECISION STANDARD

Use a moment only when its definition, estimator, stability, and decision relevance are documented and when a full-distribution diagnostic does not contradict the summary.

Moment notation and unit discipline

BEGINNER

SYMBOL	TYPE	MEANING
X, x_i	variable / realization	Uncertain outcome and observed values
$\mu, \bar{x}$	means	Population expectation and sample average
σ^2, s^2	variance	Population quantity and sample estimate
σ, s	standard deviation	Spread in original units
m_k'	raw moment	Average of X raised to power k
μ_k	central moment	Average k th power around the mean
γ_1	skewness	Standardized third central moment
β_2	kurtosis	Standardized fourth central moment
β_{2-3}	excess kurtosis	Normal-benchmark convention
n, n_{eff}	sample sizes	Row count and effective information

UNIT LADDER

Mean and standard deviation use X units; variance uses squared units; third and fourth central moments use cubed and fourth-power units. Skewness and kurtosis are unitless after standardization.

CONVENTION GATE

Always label population versus sample, divisor or bias correction, weighting, missing policy, window, horizon, and whether kurtosis is Pearson or excess. These choices are part of the metric definition.

Moment estimation workflow

BEGINNER

1

Define X

Outcome, unit, horizon, clock, population

2

Audit observations

Missingness, staleness, outliers, adjustments

3

Choose estimator

Sample, weighted, rolling, robust, conditional

4

Check assumptions

Existence, dependence, stationarity, support

5

Calculate safely

Stable algorithms and explicit conventions

6

Diagnose influence

Contributions, tails, regimes, resamples

7

Validate in time

Untouched windows and online replay

8

Publish and monitor

Lineage, uncertainty, drift, rollback

HARD GATE

No moment reaches a decision system until its variable, estimator convention, information cutoff, influence profile, and stability evidence are reproducible.

Expected value as probability-weighted center

BEGINNER

ONE-SENTENCE MEANING

Expected value is the probability-weighted average outcome of a fully specified random variable. It describes a distributional center, not a promised result or necessarily its most likely value.

MEMORY JOGGER

Imagine repeating the same well-defined experiment many times under stable probabilities; the long-run average tends toward the expectation when the required conditions hold.

WHY IT MATTERS IN QUANT RESEARCH

Expected return, transaction cost, slippage, payoff, and utility let researchers compare uncertain alternatives on a common scale. The number is only meaningful after horizon, units, conditioning information, and population are fixed.

FORMULA INTUITION

$E[X] = \sum x \times p(x)$, or $E[X] = \int x f(x) dx$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A trade pays $+2R$ with probability 0.45 and $-1R$ with probability 0.55. Gross expectation is $0.35R$, but a $0.12R$ expected cost reduces the decision-relevant value to $0.23R$.

PYTHON / SYSTEM IMPLEMENTATION

Represent scenarios and probabilities in aligned arrays, validate nonnegative weights that sum to one, and retain state-level contribution $x \cdot p(x)$. Cross-check analytic results with a seeded simulation.

CONFUSIONS TO AVOID

A positive mean can coexist with frequent losses, deep drawdowns, or a negative median. Never treat expectation as a forecast interval or as proof that one realization will be profitable.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The research ledger stores the variable contract and expectation decomposition; the strategy-ranking service consumes net expectation only after costs and uncertainty are attached.

RELATED CONCEPTS AND QUICK RECALL

Related: mean, probability weights, payoff distribution. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Discrete expectation and scenario tables

BEGINNER

ONE-SENTENCE MEANING

For a discrete variable, expectation adds each possible value multiplied by its probability mass. A complete scenario table makes the economic source of the mean auditable.

MEMORY JOGGER

Outcome times chance, row by row, then add. The contribution column matters more than a single final number because it reveals which states create the edge.

WHY IT MATTERS IN QUANT RESEARCH

Discrete scenario trees are natural for fills, gap events, earnings outcomes, defaults, and rule-based trade payoffs. They force rare but material states into the analysis.

FORMULA INTUITION

$E[X] = \sum_i x_i p_i$, with $\sum_i p_i = 1$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

For P&L states $\{-3, -0.5, 1, 4\}$ and probabilities $\{0.05, 0.35, 0.45, 0.15\}$, contributions are $\{-0.15, -0.175, 0.45, 0.60\}$; expected P&L is 0.725 units.

PYTHON / SYSTEM IMPLEMENTATION

Build a tidy table with state_id, payoff, probability, source, and as-of time. Add tests for duplicate states, missing probability mass, impossible payoffs, and sensitivity to each probability.

CONFUSIONS TO AVOID

Omitting an unobserved crash state and renormalizing the remaining rows inflates expectation. Empirical frequencies are estimates, not known probabilities, especially for rare states.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The scenario service publishes both aggregate expectation and contribution tables so stress testing can replace probabilities or payoffs without rebuilding unrelated logic.

RELATED CONCEPTS AND QUICK RECALL

Related: PMF, scenario analysis, contribution attribution. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Continuous expectation and integration

BEGINNER

ONE-SENTENCE MEANING

For a continuous variable, expectation is the integral of value times density across the support. Probability is area, so the density and measurement units must be interpreted together.

MEMORY JOGGER

Replace the discrete weighted sum with infinitely fine weighted slices. Check whether the integral converges before reporting a mean.

WHY IT MATTERS IN QUANT RESEARCH

Continuous expectations describe returns, duration, spreads, impact, and model residuals. Numerical integration is often needed when a payoff is nonlinear or a fitted distribution has no convenient closed form.

FORMULA INTUITION

$E[g(X)] = \int g(x) f_X(x) dx$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

If X is a normal daily return with mean 0.04%, then $E[X]=0.04\%$ regardless of volatility. For an option-like payoff $\max(X-K,0)$, the entire density above K affects expectation.

PYTHON / SYSTEM IMPLEMENTATION

Use stable quadrature on declared bounds, handle infinite support with library routines, and compare the result with Monte Carlo plus a standard error. Preserve whether returns are decimals or percentage points.

CONFUSIONS TO AVOID

Integrating density height without multiplying by dx is not probability. Truncating an infinite integral at convenient bounds can silently discard economically important tail contribution.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The distribution library exposes expectation of registered payoff functions and records integration tolerances, support bounds, parameters, and numerical warnings with each result.

RELATED CONCEPTS AND QUICK RECALL

Related: PDF, numerical quadrature, nonlinear payoff. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Linearity of expectation

BEGINNER

ONE-SENTENCE MEANING

The expectation of a sum equals the sum of expectations, whether or not the variables are independent. Constants pass through and scalar multipliers scale the mean directly.

MEMORY JOGGER

Means add without an independence assumption. Dependence becomes essential for variance and tail risk, but not for the expected total itself.

WHY IT MATTERS IN QUANT RESEARCH

Linearity makes portfolio expected return, total expected cost, and multi-leg payoff attribution tractable. It also lets system owners reconcile an aggregate mean to component means.

FORMULA INTUITION

$E[aX + bY + c] = aE[X] + bE[Y] + c$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A portfolio with weights 0.6 and 0.4 and expected returns 0.08% and 0.03% has expected return 0.06% per period, before financing and trading costs.

PYTHON / SYSTEM IMPLEMENTATION

Compute component contribution $w_i \mu_i$ in a table and assert that contributions sum to the portfolio result within tolerance. Keep weight and forecast timestamps aligned.

CONFUSIONS TO AVOID

Using linearity to add volatilities is wrong. Correlated positions can have an additive expected return while their combined risk depends strongly on covariance.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The portfolio forecaster assembles position-level means into an aggregate forecast; the risk engine separately supplies dependence-aware dispersion and tail estimates.

RELATED CONCEPTS AND QUICK RECALL

Related: portfolio mean, covariance, contribution analysis. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Expected payoff versus expected return

BEGINNER

ONE-SENTENCE MEANING

Expected payoff is measured in currency or payoff units, while expected return scales gain or loss by capital, price, or risk. The denominator determines economic comparability.

MEMORY JOGGER

A dollar edge and a percentage edge answer different questions. Always name the capital base, exposure convention, and holding period.

WHY IT MATTERS IN QUANT RESEARCH

Strategies with different notional, margin, and duration cannot be ranked by raw payoff. Converting to a consistent return or risk-adjusted objective prevents scale from masquerading as quality.

FORMULA INTUITION

$E[R] = E[P\&L / \text{capital_base}]$, when the base is known at decision time. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

Two trades each have expected profit of \$100, but one ties up \$5,000 and the other \$50,000. Their expected returns are 2% and 0.2% before accounting for duration and risk.

PYTHON / SYSTEM IMPLEMENTATION

Store payoff, gross exposure, net liquidation value, margin, and chosen denominator as separate fields. Reject records whose denominator is zero, future-known, or inconsistent across assets.

CONFUSIONS TO AVOID

Dividing expected P&L by average realized capital can introduce look-ahead and nonlinear bias. Leverage also changes liquidation and financing risk beyond a simple rescaling.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The opportunity-ranking layer normalizes forecast payoffs using a versioned capital convention before allocation compares opportunities across instruments.

RELATED CONCEPTS AND QUICK RECALL

Related: P&L, capital efficiency, leverage. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Costs and net expectation

BEGINNER

ONE-SENTENCE MEANING

Net expectation subtracts every decision-relevant expected cost from gross expected payoff, including spread, impact, fees, borrow, financing, and adverse selection.

MEMORY JOGGER

Edge is what remains after implementation. Small gross signals are particularly vulnerable because several uncertain costs compound.

WHY IT MATTERS IN QUANT RESEARCH

A backtest can show attractive gross alpha that cannot survive realistic execution. Modeling expected cost by size, venue, volatility, and urgency is central to deciding whether a signal is tradeable.

FORMULA INTUITION

$E[\text{net P\&L}] = E[\text{gross P\&L}] - E[\text{explicit} + \text{implicit costs}]$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A forecast offers 8 bps gross. Half-spread is 1.5 bps, expected impact 2.2, fees 0.3, borrow 0.8, and slippage uncertainty adds a conservative 0.7 buffer, leaving 2.5 bps.

PYTHON / SYSTEM IMPLEMENTATION

Calculate costs from point-in-time quotes and order attributes, then version the model. Log predicted and realized cost by component so calibration can be monitored without contaminating old tests.

CONFUSIONS TO AVOID

Applying today's fee schedule or liquidity model to historical orders is look-ahead. Ignoring cost variance also overstates confidence even when average net expectation stays positive.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The execution simulator returns a cost distribution to the signal evaluator; the go/no-go rule consumes net expectation and uncertainty rather than a fixed haircut.

RELATED CONCEPTS AND QUICK RECALL

Related: transaction costs, market impact, implementation shortfall. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Conditional expectation

BEGINNER

ONE-SENTENCE MEANING

Conditional expectation is the mean of X after restricting information to a known event, variable, or information set. It is itself a function of what is observed.

MEMORY JOGGER

Unconditional mean blends states; conditional mean asks what is expected now, given the information available now.

WHY IT MATTERS IN QUANT RESEARCH

Signals forecast conditional returns, not timeless averages. Regime, volatility, inventory, spread, and event context can all shift the relevant expectation.

FORMULA INTUITION

$m(z) = E[X | Z=z]$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

Average return may be near zero, while $E[R_{\text{next}} | \text{high volatility, negative trend}]$ is materially negative. That conditional estimate still requires enough observations in the specified state.

PYTHON / SYSTEM IMPLEMENTATION

Define feature timestamps and prediction cutoff explicitly. Fit $m(z)$ only on past data, retain coverage by state, and compare conditional forecasts with an unconditional benchmark out of sample.

CONFUSIONS TO AVOID

Conditioning on a future label, revised macro value, or end-of-day statistic creates leakage. Very narrow conditioning sets can also turn noise into extreme apparent means.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature store enforces availability time; the forecast service records the information set used for every conditional mean emitted before the outcome occurs.

RELATED CONCEPTS AND QUICK RECALL

Related: regression function, information set, regime. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Law of iterated expectations

BEGINNER

ONE-SENTENCE MEANING

The unconditional expectation equals the expectation of conditional expectations across conditioning states. This tower property connects segmented forecasts to the overall mean.

MEMORY JOGGER

Average the within-state means using state probabilities and you recover the total mean, provided the partition and weights are consistent.

WHY IT MATTERS IN QUANT RESEARCH

The identity supports forecast attribution, mixture models, and reconciliation between regime dashboards and aggregate expectations. It also catches inconsistent samples or weighting.

FORMULA INTUITION

$E[X] = E[E[X | Z]]$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

If calm and stress states have probabilities 0.8 and 0.2 with expected returns 0.05% and -0.30%, total expectation is $0.8(0.05\%) + 0.2(-0.30\%) = -0.02\%$.

PYTHON / SYSTEM IMPLEMENTATION

Implement a reconciliation test that compares the direct aggregate mean with weighted group means using the same rows and weights. Fail when coverage or missing-state treatment differs.

CONFUSIONS TO AVOID

Averaging group means equally instead of by state probability changes the target population. Overlapping groups cannot be treated as a simple partition without adjusted weights.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The monitoring dashboard reconciles regime-conditioned forecasts to the global forecast and highlights whether changes come from state frequency or within-state mean.

RELATED CONCEPTS AND QUICK RECALL

Related: tower property, mixtures, decomposition. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Jensen's inequality and nonlinear transforms

BEGINNER

ONE-SENTENCE MEANING

For a convex function, the function of the mean is no greater than the mean of the function. Nonlinearity therefore makes plug-in averages systematically misleading.

MEMORY JOGGER

Average-then-transform usually differs from transform-then-average. Curvature decides the direction of the gap.

WHY IT MATTERS IN QUANT RESEARCH

Compounding, utility, option payoffs, log transforms, volatility drag, and exponentiated forecasts all involve nonlinear functions. Jensen's gap is often economically material.

FORMULA INTUITION

Convex g : $g(E[X]) \leq E[g(X)]$; concave g reverses the inequality. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

If log return Y is normal with mean 0.01 and variance 0.04, $E[\exp(Y)] = \exp(0.01+0.02)$, not $\exp(0.01)$. Variance raises the expected gross price ratio.

PYTHON / SYSTEM IMPLEMENTATION

Whenever inverse-transforming a forecast, simulate or use the analytic distributional correction. Unit-test against a degenerate zero-variance case where both sides coincide.

CONFUSIONS TO AVOID

Exponentiating an average log return estimates the median gross ratio under lognormality, not the arithmetic mean. The correction also depends on a credible variance estimate.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The transformation registry marks nonlinear inverse maps and requires a bias-correction method before transformed predictions can reach reporting or allocation.

RELATED CONCEPTS AND QUICK RECALL

Related: convexity, lognormal correction, compounding. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

When expectation does not exist

BEGINNER

ONE-SENTENCE MEANING

An expectation exists only when the probability-weighted magnitude is integrable. Some heavy-tailed models have undefined or unstable means even though finite samples always produce a number.

MEMORY JOGGER

A spreadsheet can calculate a sample average when the population mean is infinite. Existence is a model property, not a software output property.

WHY IT MATTERS IN QUANT RESEARCH

Tail models and extreme-payoff strategies can make mean-based ranking meaningless. Researchers must check fitted tail parameters and avoid claiming convergence unsupported by the distribution.

FORMULA INTUITION

$E[|X|] < \infty$ is the usual existence condition. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A Pareto variable with tail index $\alpha \leq 1$ has no finite mean. A simulation may show a plausible average, but it will jump as rare larger observations arrive.

PYTHON / SYSTEM IMPLEMENTATION

Expose a `moments_available` method for each distribution family and parameter set. In simulation studies, plot running means and maxima across increasing sample sizes and independent seeds.

CONFUSIONS TO AVOID

Winsorizing data makes the transformed sample mean finite but does not prove the original population expectation exists. It changes the estimand and must be labeled.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The model registry blocks downstream mean consumers when existence conditions fail and routes the decision toward quantiles, bounded utility, or explicitly truncated scenarios.

RELATED CONCEPTS AND QUICK RECALL

Related: tail index, integrability, robust summaries. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Variance as expected squared deviation

BEGINNER

ONE-SENTENCE MEANING

Variance measures the expected squared distance from the mean. Squaring prevents positive and negative deviations from canceling and makes large departures disproportionately influential.

MEMORY JOGGER

Center, square, then average. Variance describes spread in squared units; standard deviation converts that spread back to the original unit.

WHY IT MATTERS IN QUANT RESEARCH

Variance underpins volatility forecasts, tracking error, forecast uncertainty, covariance matrices, and many portfolio objectives. It summarizes dispersion but not direction or full tail shape.

FORMULA INTUITION

$\text{Var}(X) = E[(X - \mu)^2]$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

For outcomes $\{-1, 2\}$ with probabilities $\{0.4, 0.6\}$, $\mu = 0.8$. Squared deviations are 3.24 and 1.44, so variance is $0.4(3.24) + 0.6(1.44) = 2.16$.

PYTHON / SYSTEM IMPLEMENTATION

Use a stable two-pass calculation or a tested online algorithm. Preserve units, horizon, missing policy, weights, and degrees of freedom with the result.

CONFUSIONS TO AVOID

Variance treats upside and downside equally, is highly sensitive to extremes, and may not exist for heavy-tailed distributions. It is not automatically a loss measure.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The risk feature service computes point-in-time variance estimates that feed portfolio risk, volatility scaling, and monitoring under a shared estimator contract.

RELATED CONCEPTS AND QUICK RECALL

Related: dispersion, standard deviation, second central moment. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Computational identity for variance

BEGINNER

ONE-SENTENCE MEANING

Variance equals the second raw moment minus the square of the mean. The identity is algebraically useful but can be numerically unstable when large terms nearly cancel.

MEMORY JOGGER

Average squares, subtract squared average. Use the formula for reasoning; use stable algorithms for floating-point computation.

WHY IT MATTERS IN QUANT RESEARCH

The identity enables analytic derivations and moment decomposition. It also reveals that changing the mean changes variance even if the raw second moment stays fixed.

FORMULA INTUITION

$\text{Var}(X) = E[X^2] - (E[X])^2$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

For X in $\{-1, 2\}$ with probabilities $\{0.4, 0.6\}$, $E[X^2]=2.8$ and $E[X]^2=0.64$, giving variance 2.16.

PYTHON / SYSTEM IMPLEMENTATION

Test the identity on well-scaled toy data, but compute production variance with centered values, pairwise aggregation, or Welford's algorithm. Compare against high-precision references.

CONFUSIONS TO AVOID

For values near one billion with tiny deviations, subtracting two roughly 10^{18} quantities can lose the small variance through catastrophic cancellation.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The statistics library centralizes stable implementations so notebooks, batch jobs, and streaming monitors do not invent inconsistent variance calculations.

RELATED CONCEPTS AND QUICK RECALL

Related: raw moment, cancellation, numerical stability. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Population variance and sample variance

BEGINNER

ONE-SENTENCE MEANING

Population variance describes a distribution, while sample variance estimates it from observed data. Dividing by $n-1$ corrects finite-sample bias under the standard IID setting.

MEMORY JOGGER

Use n for the descriptive variance of the rows in hand; use $n-1$ for the familiar unbiased estimator of an unknown IID population variance.

WHY IT MATTERS IN QUANT RESEARCH

Research code must distinguish descriptive backtest dispersion from statistical estimation. The convention affects volatility, standard errors, and reproducibility, especially in small samples.

FORMULA INTUITION

$s^2 = \sum_i (x_i - \bar{x})^2 / (n-1)$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

For returns $\{1\%, 2\%, 3\%$, $\bar{x}=2\%$ and squared deviations total 2 percentage-points squared. Population divisor gives 0.667; sample divisor gives 1.0.

PYTHON / SYSTEM IMPLEMENTATION

Pass `ddof` explicitly in NumPy or pandas and label the output. Reject $n < 2$ for sample variance and record whether observations are weighted or dependent.

CONFUSIONS TO AVOID

Bessel's correction does not solve autocorrelation, heteroskedasticity, selection bias, or nonstationarity. Unbiasedness under an ideal model is not general reliability.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The metric catalog defines divisor conventions so backtest summaries, risk reports, and model features agree across languages and services.

RELATED CONCEPTS AND QUICK RECALL

Related: Bessel correction, estimator bias, degrees of freedom. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Standard deviation and volatility units

BEGINNER

ONE-SENTENCE MEANING

Standard deviation is the square root of variance and therefore uses the same units as the underlying variable. In finance, return standard deviation over a stated horizon is commonly called volatility.

MEMORY JOGGER

Variance is mathematically convenient; standard deviation is easier to interpret because it returns to dollars, returns, basis points, or seconds.

WHY IT MATTERS IN QUANT RESEARCH

Volatility supports risk limits, normalization, position sizing, forecast intervals, and cross-asset comparison. Horizon and estimator must accompany every value.

FORMULA INTUITION

$\sigma = \sqrt{\text{Var}(X)}$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

If daily return variance is 0.000225, daily standard deviation is 0.015, or 1.5%. This does not mean returns stay within $\pm 1.5\%$ or follow a normal distribution.

PYTHON / SYSTEM IMPLEMENTATION

Store volatility as a decimal with a horizon enum and estimator ID. Format percentages only at presentation boundaries and test decimal-versus-percent conversions.

CONFUSIONS TO AVOID

Calling standard deviation a typical maximum move is wrong. A 2-sigma interpretation requires distributional assumptions, while realized volatility depends on sampling and microstructure choices.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The risk API transports unit-safe volatility objects to sizing and limits; presentation code cannot silently annualize or rescale them.

RELATED CONCEPTS AND QUICK RECALL

Related: volatility, units, forecast interval. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Variance under scaling and translation

BEGINNER

ONE-SENTENCE MEANING

Adding a constant shifts the mean but leaves variance unchanged; multiplying by a constant multiplies variance by the constant squared.

MEMORY JOGGER

Location does not change spread. Scale changes deviations linearly and squared deviations quadratically.

WHY IT MATTERS IN QUANT RESEARCH

The rule explains leverage effects, currency conversion, contract multipliers, and standardized features. It is also a basic unit test for risk engines.

FORMULA INTUITION

$\text{Var}(aX+b) = a^2 \text{Var}(X)$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

Doubling a position doubles each P&L outcome but quadruples P&L variance. Adding a fixed \$10 rebate to every outcome changes expected P&L by \$10 without changing variance.

PYTHON / SYSTEM IMPLEMENTATION

Write property-based tests using random arrays: translated variance should match, while scaled variance should change by a^2 within tolerance. Verify negative a as well.

CONFUSIONS TO AVOID

Confusing variance scaling with standard-deviation scaling leads to leverage errors. Standard deviation changes by $|a|$, not a^2 .

WHERE THIS FITS IN THE RESEARCH SYSTEM

The instrument-normalization layer applies contract multipliers before the risk engine aggregates P&L variance across positions.

RELATED CONCEPTS AND QUICK RECALL

Related: location-scale transform, leverage, dimensional analysis. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Variance of sums and covariance

BEGINNER

ONE-SENTENCE MEANING

Variance of a sum includes each component variance plus covariance terms. Independence removes covariance, but uncorrelatedness is the exact condition needed for the cross terms to vanish.

MEMORY JOGGER

Means always add; risks add only after accounting for co-movement.

WHY IT MATTERS IN QUANT RESEARCH

Portfolio risk, spread risk, hedging, and factor aggregation depend on covariance. Ignoring it can either overstate diversification or miss concentration.

FORMULA INTUITION

$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

Two assets each have variance 4 and covariance 3. Equal one-unit positions have total variance $4+4+6=14$, not 8.

PYTHON / SYSTEM IMPLEMENTATION

Align timestamps and missingness before covariance estimation. Reconstruct aggregate variance from component and pair contributions and assert agreement with direct P&L variance.

CONFUSIONS TO AVOID

Adding individual variances assumes zero covariance. Correlation near zero in one window does not guarantee stable diversification during stress.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The covariance estimator supplies a positive-semidefinite matrix and contribution breakdown to portfolio construction and stress testing.

RELATED CONCEPTS AND QUICK RECALL

Related: covariance, correlation, diversification. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Portfolio variance and marginal risk

BEGINNER

ONE-SENTENCE MEANING

Portfolio variance is a quadratic form in weights and the covariance matrix. Marginal risk measures how a small weight change affects total variance or volatility.

MEMORY JOGGER

Weights enter twice because every position can co-move with every other position.

WHY IT MATTERS IN QUANT RESEARCH

The quadratic form is the core of mean-variance allocation, risk budgeting, hedge analysis, and pre-trade limit checks.

FORMULA INTUITION

$\text{Var}(r_p) = w^T \Sigma w$; gradient = $2 \Sigma w$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

With two assets, weights $\{0.6, 0.4\}$, volatilities $\{10\%, 20\%\}$, and correlation 0.25, variance is $0.0036 + 0.0064 + 0.0024 = 0.0124$, so volatility is 11.14%.

PYTHON / SYSTEM IMPLEMENTATION

Validate matrix symmetry, feature order, units, and positive semidefiniteness. Compare quadratic-form output with scenario P&L variance on a controlled dataset.

CONFUSIONS TO AVOID

A mathematically valid covariance matrix can still be stale or poorly estimated. Weight constraints and transaction costs may dominate small theoretical risk improvements.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The allocation engine consumes timestamped forecasts and covariance artifacts, while the risk service recomputes marginal and component contributions after every proposed trade.

RELATED CONCEPTS AND QUICK RECALL

Related: quadratic form, risk contribution, optimization. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Law of total variance

BEGINNER

ONE-SENTENCE MEANING

Total variance equals expected within-state variance plus variance of state-conditioned means. It separates ordinary noise from dispersion caused by changing regimes.

MEMORY JOGGER

Overall uncertainty = average uncertainty inside states + uncertainty about which state and state mean applies.

WHY IT MATTERS IN QUANT RESEARCH

The decomposition shows whether risk comes from volatile behavior within regimes or from shifts between calm, trend, and stress states. Those sources require different controls.

FORMULA INTUITION

$\text{Var}(X) = E[\text{Var}(X|Z)] + \text{Var}(E[X|Z])$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

Two equally likely regimes have means -1 and +1 and within-regime variance 4. Expected within variance is 4; variance of regime means is 1; total variance is 5.

PYTHON / SYSTEM IMPLEMENTATION

Compute both terms from one consistent partition and reconcile to direct variance. Preserve state probabilities, conditional means, variances, and residual differences.

CONFUSIONS TO AVOID

Using future-defined regimes makes the decomposition descriptive but not deployable. Noisy state labels can also exaggerate between-state variation.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The risk dashboard attributes variance to within-regime turbulence and regime switching, guiding whether to reduce exposure or improve state estimation.

RELATED CONCEPTS AND QUICK RECALL

Related: conditional variance, regimes, mixture decomposition. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Downside variance and semideviation

BEGINNER

ONE-SENTENCE MEANING

Downside variance measures squared shortfalls below a target, focusing dispersion on outcomes considered harmful. Semideviation is its square root.

MEMORY JOGGER

Do not penalize all movement when only misses below a threshold matter; define the threshold before observing results.

WHY IT MATTERS IN QUANT RESEARCH

Downside measures are useful for capital preservation, target-return mandates, and asymmetric strategies where upside variability is not a risk in the same sense as loss.

FORMULA INTUITION

$\text{DownsideVar}_{\tau} = E[\min(X - \tau, 0)^2]$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

For returns $\{-3\%, -1\%, 2\%, 4\%$ with target 0, downside squares are $\{0.0009, 0.0001, 0, 0\}$; their average is 0.00025 and semideviation is 1.58%.

PYTHON / SYSTEM IMPLEMENTATION

Require a target, horizon, denominator convention, and missing policy. Report downside observation count and compare with ordinary variance and expected shortfall.

CONFUSIONS TO AVOID

Different libraries divide by total n or only downside count, producing different values. Downside variance still squares losses and can hide tail shape.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The strategy scorecard uses a cataloged downside estimator for asymmetric risk objectives while the core risk engine retains full-distribution scenarios.

RELATED CONCEPTS AND QUICK RECALL

Related: lower partial moment, Sortino ratio, target return. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Variance estimation uncertainty

BEGINNER

ONE-SENTENCE MEANING

A variance estimate is itself uncertain and often much noisier than its point display suggests. Dependence, volatility clustering, and heavy tails reduce effective information.

MEMORY JOGGER

Risk has estimation risk. More rows do not equal more independent evidence when observations cluster through time or share factors.

WHY IT MATTERS IN QUANT RESEARCH

Position sizing and optimization can react aggressively to noisy covariance estimates. Confidence ranges and stability diagnostics prevent false precision.

FORMULA INTUITION

Under IID normality, $(n-1)s^2/\sigma^2$ follows $\text{chi-square}_{(n-1)}$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

With only 20 IID normal observations, the variance confidence interval is wide. Financial returns usually violate the ideal assumptions, widening practical uncertainty further.

PYTHON / SYSTEM IMPLEMENTATION

Use block bootstrap or model-based simulation when dependence matters. Track estimate dispersion across windows, half-lives, and resamples rather than publishing only one number.

CONFUSIONS TO AVOID

The textbook chi-square interval is fragile under heavy tails and autocorrelation. A narrow decimal format is not evidence of precision.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The risk governance layer attaches uncertainty and stability flags to variance artifacts before optimization is allowed to rely on them.

RELATED CONCEPTS AND QUICK RECALL

Related: confidence interval, effective sample size, bootstrap. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Raw, central, and standardized moments

BEGINNER

ONE-SENTENCE MEANING

Raw moments average powers of X , central moments average powers of deviations from the mean, and standardized moments divide by the appropriate power of standard deviation.

MEMORY JOGGER

Raw moments depend on location; central moments describe shape around the center; standardization removes measurement scale.

WHY IT MATTERS IN QUANT RESEARCH

This hierarchy organizes mean, variance, skewness, and kurtosis and makes their units clear. It also exposes which distributional summaries may fail to exist.

FORMULA INTUITION

raw $m_k = E[X^k]$; central $\mu_k = E[(X - \mu)^k]$; standardized μ_k / σ^k . Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

For $k=1$ the first central moment is zero. For $k=2$ the standardized central moment is one. Shape information begins with standardized orders three and four.

PYTHON / SYSTEM IMPLEMENTATION

Implement one tested moments function with explicit centering, scaling, weights, and bias convention. Return metadata rather than unlabeled scalar arrays.

CONFUSIONS TO AVOID

Comparing unstandardized third or fourth moments across assets mostly compares scale. Higher powers amplify outliers and floating-point problems.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature library provides versioned moment definitions so research notebooks, monitoring, and model diagnostics share identical conventions.

RELATED CONCEPTS AND QUICK RECALL

Related: moment order, centralization, standardization. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Skewness as directional asymmetry

BEGINNER

ONE-SENTENCE MEANING

Skewness is the standardized third central moment. Its sign indicates which side has the longer or more influential tail, while magnitude summarizes asymmetry imperfectly.

MEMORY JOGGER

Cubed deviations preserve sign: large right-side observations push skew positive; large left-side observations push it negative.

WHY IT MATTERS IN QUANT RESEARCH

Skewness distinguishes strategies with similar mean and volatility but different crash or windfall tendencies. It matters for options, carry, trend, and stop-based payoffs.

FORMULA INTUITION

$\gamma_1 = E[(X - \mu)^3] / \sigma^3$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A short-volatility strategy may earn frequent small gains and occasional large losses, producing negative skew even when its average return is positive.

PYTHON / SYSTEM IMPLEMENTATION

Calculate skew with a documented finite-sample convention, show the return histogram or ECDF, and report sensitivity to influential dates and rolling windows.

CONFUSIONS TO AVOID

Positive skew does not mean most observations are positive, and negative skew does not identify a particular tail probability. Skewness can be driven by a single event.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The strategy diagnostics service pairs skewness with quantiles, drawdowns, expected shortfall, and event attribution before risk review.

RELATED CONCEPTS AND QUICK RECALL

Related: third central moment, asymmetry, tail loss. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Sample skewness and bias correction

BEGINNER

ONE-SENTENCE MEANING

Sample skewness estimates population asymmetry from finite data. Several formulas differ in small-sample bias correction, so software outputs can disagree without either being broken.

MEMORY JOGGER

The basic standardized third moment is descriptive; corrected versions target an estimator under additional assumptions.

WHY IT MATTERS IN QUANT RESEARCH

Small trade counts and short histories make skew estimates unstable. Reproducible research requires recording the exact convention and sample eligibility.

FORMULA INTUITION

$G1 = [\text{sqrt}(n(n-1)) / (n-2)] * m3 / m2^{(3/2)}$, for $n > 2$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

For $n=10$, the correction factor is materially above one. As n grows, the corrected and uncorrected coefficients approach each other.

PYTHON / SYSTEM IMPLEMENTATION

Set `bias=False` or the equivalent explicitly, verify against a hand calculation, and return n plus missing-count metadata. Add a minimum-sample rule.

CONFUSIONS TO AVOID

Bias correction cannot repair dependence, regime change, or infinite third moments. Reporting corrected skew to four decimals still implies false precision.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The metric registry includes formula ID and library version so historical scorecards remain comparable after code upgrades.

RELATED CONCEPTS AND QUICK RECALL

Related: estimator convention, finite sample, reproducibility. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Interpreting positive and negative skew

BEGINNER

ONE-SENTENCE MEANING

Positive skew reflects greater influence from large positive deviations, while negative skew reflects greater influence from large negative deviations. The sign is about centered cubic contribution, not win rate.

MEMORY JOGGER

Ask which observations dominate the cubed deviations, then inspect their economic mechanism.

WHY IT MATTERS IN QUANT RESEARCH

Payoff shape matters: lottery-like strategies often have positive skew; premium-selling and carry strategies often have negative skew. Investors may value these profiles differently.

FORMULA INTUITION

$\text{sign}(\gamma_1) = \text{sign}(E[(X-\mu)^3])$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A strategy wins 70% of days but loses 8% in rare shocks; it can have negative skew. Another loses slightly most days but occasionally gains 15%; it can have positive skew.

PYTHON / SYSTEM IMPLEMENTATION

Produce a contribution table $(x_i - \bar{x})^3$, sorted by absolute contribution. Link the largest observations to market events and data-quality checks.

CONFUSIONS TO AVOID

Win rate, median, and skew sign are separate properties. Clipping or stop rules can change skew while also changing mean and dependence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The research review UI links moment contributions to event logs, enabling the analyst to distinguish real payoff asymmetry from bad ticks.

RELATED CONCEPTS AND QUICK RECALL

Related: win rate, payoff convexity, event attribution. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Kurtosis as standardized fourth moment

BEGINNER

ONE-SENTENCE MEANING

Kurtosis is the standardized fourth central moment. It measures the influence of extreme deviations and overall distributional shape, not simply how sharp the center looks.

MEMORY JOGGER

Fourth powers make both tails positive and strongly emphasize distant observations.

WHY IT MATTERS IN QUANT RESEARCH

High kurtosis warns that variance may be carried by rare extremes and that Gaussian risk approximations can understate event frequency. It is common in financial returns.

FORMULA INTUITION

$\text{beta}_2 = E[(X-\mu)^4] / \sigma^4$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

One -10% return contributes 10,000 times as much fourth-power deviation as a -1% return before standardization, showing why kurtosis is extremely event-sensitive.

PYTHON / SYSTEM IMPLEMENTATION

Report kurtosis alongside sample size, largest fourth-moment contributors, tail quantiles, and an estimator convention. Stress the result by removing one observation at a time.

CONFUSIONS TO AVOID

Kurtosis alone does not reveal which tail is dangerous or the exact probability of a loss threshold. Different shapes can share the same kurtosis.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The distribution monitor uses kurtosis as a trigger for deeper tail diagnostics, not as a standalone limit or model-selection score.

RELATED CONCEPTS AND QUICK RECALL

Related: fourth central moment, tail influence, shape. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Excess kurtosis and the normal benchmark

BEGINNER

ONE-SENTENCE MEANING

Excess kurtosis subtracts three so a normal distribution has value zero. Positive excess indicates more fourth-moment weight than normal; negative excess indicates less.

MEMORY JOGGER

Raw kurtosis of normal = 3; excess kurtosis of normal = 0. Always state which convention is reported.

WHY IT MATTERS IN QUANT RESEARCH

The normal benchmark makes dashboards easier to read, but a zero estimate does not prove normality. Other distributions and finite samples can also produce values near zero.

FORMULA INTUITION

$\text{excess kurtosis} = \text{beta}_2 - 3$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

If software reports Fisher kurtosis 5, excess is 5; if it reports Pearson kurtosis, the same sample is shown as 8.

PYTHON / SYSTEM IMPLEMENTATION

Name fields `kurtosis_excess` or `kurtosis_pearson` rather than `kurtosis`. Add a unit test using a large seeded normal sample and library-specific tolerances.

CONFUSIONS TO AVOID

Comparing a Fisher output with a Pearson output creates a false three-point discrepancy. Normal-like kurtosis does not imply normal skew, tails, or dependence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The semantic metric catalog prevents convention drift across Python, SQL, and reporting tools by encoding the benchmark explicitly.

RELATED CONCEPTS AND QUICK RECALL

Related: Fisher convention, Pearson convention, normal distribution. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Kurtosis, tails, and center shape

BEGINNER

ONE-SENTENCE MEANING

Kurtosis depends on the whole standardized distribution and can reflect tail weight, shoulders, or a mixture structure. It should not be reduced to a claim about peak height.

MEMORY JOGGER

Fourth-moment mass can come from rare tails or from how probability is arranged between center and extremes.

WHY IT MATTERS IN QUANT RESEARCH

Returns often mix quiet and stressed regimes, producing high kurtosis even when each regime is relatively ordinary. Understanding the mechanism matters for scenarios and controls.

FORMULA INTUITION

$\beta_2 = \int \frac{(x-\mu)^4}{\sigma^4} f(x) dx$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A two-volatility mixture with the same mean can show high kurtosis because occasional high-variance states generate far more extreme standardized outcomes.

PYTHON / SYSTEM IMPLEMENTATION

Compare full-sample and regime-conditioned distributions, plot standardized ECDF tails, and decompose the fourth moment by return bins and regimes.

CONFUSIONS TO AVOID

Calling kurtosis 'peakedness' can misdiagnose a mixture. A narrow center and heavy tails may coexist, but one does not mechanically cause the other.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The regime modeling pipeline tests whether excess kurtosis falls after conditioning, which indicates that state mixing explains part of the aggregate shape.

RELATED CONCEPTS AND QUICK RECALL

Related: mixture distribution, regime risk, tail weight. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Heavy tails and moment existence

BEGINNER

ONE-SENTENCE MEANING

A distribution can have a finite mean but infinite variance, or finite variance but infinite skewness or kurtosis. Higher moments require progressively stronger tail decay.

MEMORY JOGGER

Before trusting order k , ask whether the tail is light enough for $E[|X|^k]$ to exist.

WHY IT MATTERS IN QUANT RESEARCH

Many financial tail models imply that sample higher moments converge slowly or not at all. This limits Gaussian approximations and moment-based optimization.

FORMULA INTUITION

For Pareto tail index α , moment k exists only when $k < \alpha$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

If $\alpha=3.5$, mean, variance, and third moment exist, but the fourth moment does not. Sample kurtosis will remain unstable as the sample grows.

PYTHON / SYSTEM IMPLEMENTATION

Estimate tail behavior with uncertainty, inspect running moments across sample size, and disable unavailable analytic moments in model objects.

CONFUSIONS TO AVOID

A finite sample always returns finite skewness and kurtosis unless numerically broken. That fact does not establish corresponding population moments.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The tail-risk gate blocks moment-dependent models when fitted tail evidence does not support the required moment order.

RELATED CONCEPTS AND QUICK RECALL

Related: Pareto tail index, moment existence, convergence. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Outlier sensitivity and influence

BEGINNER

ONE-SENTENCE MEANING

Higher moments give extreme observations enormous leverage because deviations are raised to the third and fourth powers. Influence analysis is therefore mandatory.

MEMORY JOGGER

An observation twice as far from the mean has eight times the cubic contribution and sixteen times the fourth-power contribution.

WHY IT MATTERS IN QUANT RESEARCH

A bad tick, split error, stale quote, or genuine crash can dominate skewness and kurtosis. The correct response depends on data provenance, not cosmetic trimming.

FORMULA INTUITION

$\text{contribution}_k(i) = (x_i - \bar{x})^k / n$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A -12% observation contributes 16 times the fourth-power mass of a -6% observation before accounting for the mean and scale.

PYTHON / SYSTEM IMPLEMENTATION

Rank contribution magnitude, inspect raw records and corporate actions, then compare original, corrected, and explicitly robust summaries. Keep an immutable audit trail.

CONFUSIONS TO AVOID

Deleting influential but valid market events makes risk look safer. Keeping known data errors is equally wrong; provenance decides treatment.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The data-quality service links flagged observations to downstream metric impact so corrections trigger controlled recomputation.

RELATED CONCEPTS AND QUICK RECALL

Related: influence function, bad ticks, data lineage. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Robust alternatives to classical moments

BEGINNER

ONE-SENTENCE MEANING

Robust summaries reduce sensitivity to extreme observations or focus on quantiles rather than powers. They trade some classical efficiency for stability and clearer tail interpretation.

MEMORY JOGGER

Median and MAD describe the center; quantiles and expected shortfall describe tails without relying on third or fourth moments.

WHY IT MATTERS IN QUANT RESEARCH

When data are heavy-tailed, contaminated, or scarce, robust alternatives can provide more stable monitoring and ranking than sample skewness or kurtosis.

FORMULA INTUITION

$MAD = \text{median}(|X - \text{median}(X)|)$; quantile skew = $(Q.75 + Q.25 - 2Q.50) / (Q.75 - Q.25)$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A single erroneous 20-sigma tick explodes classical kurtosis but has limited effect on the median and interquartile range.

PYTHON / SYSTEM IMPLEMENTATION

Compute classical and robust panels together. Version quantile interpolation and rolling-window rules, and validate decisions rather than choosing robustness solely for smoother charts.

CONFUSIONS TO AVOID

Robust does not mean immune to bias, regime change, or dependence. Winsorized moments describe a modified distribution and must not be mislabeled as original moments.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The monitoring layer uses robust statistics for alert stability while preserving untrimmed tail metrics for risk accountability.

RELATED CONCEPTS AND QUICK RECALL

Related: median absolute deviation, quantiles, winsorization. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Moment generating and characteristic functions

BEGINNER

ONE-SENTENCE MEANING

Generating functions encode distribution moments through derivatives when those moments and the function exist. Characteristic functions always exist and are especially useful for sums.

MEMORY JOGGER

A transform can package an entire distribution; derivatives at zero recover moments when justified.

WHY IT MATTERS IN QUANT RESEARCH

These tools derive aggregate distributions, connect independent sums, and support option pricing and numerical inversion. They are reference machinery rather than routine descriptive metrics.

FORMULA INTUITION

$M_X(t) = E[e^{tX}]$; $\phi_X(t) = E[e^{itX}]$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

For independent X and Y , $M_{X+Y}(t) = M_X(t)M_Y(t)$ where MGFs exist. This turns convolution into multiplication.

PYTHON / SYSTEM IMPLEMENTATION

Use symbolic or trusted numerical libraries for derivations, validate domains, and test recovered low-order moments against direct calculations or simulation.

CONFUSIONS TO AVOID

An MGF may fail to exist away from zero for heavy-tailed variables even when some moments exist. Formal differentiation without domain checks can be invalid.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The distribution library records analytic transforms only for supported families and falls back to numerical convolution or simulation elsewhere.

RELATED CONCEPTS AND QUICK RECALL

Related: MGF, characteristic function, convolution. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Moment matching and distribution fitting

BEGINNER

ONE-SENTENCE MEANING

Moment matching chooses model parameters so selected theoretical moments equal sample moments. It is simple but inherits the noise and existence problems of those moments.

MEMORY JOGGER

Choose parameters that reproduce summaries, then test whether the rest of the distribution also fits.

WHY IT MATTERS IN QUANT RESEARCH

Moment matching can initialize estimators, calibrate simple distributions, or fit scenario approximations. It is rarely sufficient for tail-sensitive decisions without held-out validation.

FORMULA INTUITION

solve $m_k(\theta) = \text{sample_}m_k$ for selected k . Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A gamma distribution can be matched using $\text{shape} = (\text{mean}/\text{std})^2$ and $\text{scale} = \text{variance}/\text{mean}$ for positive data, provided the sample summaries are credible.

PYTHON / SYSTEM IMPLEMENTATION

Check support first, solve with constraints, and compare fitted CDF, quantiles, likelihood, and tail exceedances against empirical data and alternative estimators.

CONFUSIONS TO AVOID

Matching four moments does not uniquely identify every distribution and can chase unstable sample skewness or kurtosis. Good moment fit is not good tail calibration.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The model-selection workflow treats moment matching as one challenger and stores residual diagnostics beyond the matched statistics.

RELATED CONCEPTS AND QUICK RECALL

Related: method of moments, calibration, goodness of fit. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Cumulants and additive structure

BEGINNER

ONE-SENTENCE MEANING

Cumulants re-express distribution shape so independent sums have additive cumulants. The first four correspond to mean, variance, third central moment, and excess-related fourth structure.

MEMORY JOGGER

For independent components, cumulants add more cleanly than standardized skewness or kurtosis.

WHY IT MATTERS IN QUANT RESEARCH

Cumulants help reason about aggregated independent P&L, horizon effects, and why standardized non-normality can diminish under ideal aggregation.

FORMULA INTUITION

$K_X(t) = \log M_X(t)$; $\kappa_n(X+Y) = \kappa_n(X) + \kappa_n(Y)$ for independent X, Y . Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

For n IID periods, mean and variance scale with n , while standardized skewness scales as $1/\sqrt{n}$ and excess kurtosis as $1/n$ under strong assumptions.

PYTHON / SYSTEM IMPLEMENTATION

Use cumulant identities only after checking independence and existence. Validate aggregation formulas against simulated IID and autocorrelated cases.

CONFUSIONS TO AVOID

Financial returns are not generally IID; volatility clustering and regime shifts prevent simple horizon dilution of skewness and excess kurtosis.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The scenario engine uses cumulant checks as diagnostics for aggregation assumptions, while final risk remains scenario-based when dependence is material.

RELATED CONCEPTS AND QUICK RECALL

Related: cumulant, independent sums, central limit behavior. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Annualization and square-root-of-time

BEGINNER

ONE-SENTENCE MEANING

Under independent, identically distributed increments with finite variance, expected return scales with time and variance scales linearly, so standard deviation scales with the square root of time.

MEMORY JOGGER

Mean uses T ; volatility uses $\sqrt{T}$ - only under assumptions that financial data frequently violate.

WHY IT MATTERS IN QUANT RESEARCH

Annualization allows common reporting units, but autocorrelation, time-varying volatility, overlapping returns, and compounding can invalidate mechanical scaling.

FORMULA INTUITION

$\mu_T = T \mu_1$; $\sigma_T = \sqrt{T} \sigma_1$ under IID additive returns. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

Daily volatility of 1% annualizes to about 15.87% using $\sqrt{252}$. This is a convention-based projection, not an observed annual risk guarantee.

PYTHON / SYSTEM IMPLEMENTATION

Store native-horizon statistics and annualization factor separately. Compare scaled estimates with directly measured longer-horizon returns and heteroskedastic forecasts.

CONFUSIONS TO AVOID

Applying $\sqrt{252}$ to autocorrelated strategy returns or treating 252 as universal creates false comparability. Arithmetic mean scaling ignores compounding effects.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The reporting service annualizes only through registered policies and always retains the native estimate, calendar, and diagnostic evidence.

RELATED CONCEPTS AND QUICK RECALL

Related: time aggregation, autocorrelation, compounding. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Rolling and expanding moment estimates

BEGINNER

ONE-SENTENCE MEANING

Rolling estimators use a fixed recent window; expanding estimators use all history to date. The choice trades responsiveness against sampling stability.

MEMORY JOGGER

Short windows react quickly and jump; long windows move slowly and may blend obsolete regimes.

WHY IT MATTERS IN QUANT RESEARCH

Monitoring mean, volatility, skewness, and kurtosis through time reveals regime shifts and estimator instability hidden by one full-sample statistic.

FORMULA INTUITION

$m_t = \text{statistic}(\{x_{(t-w+1)}, \dots, x_t\})$ for rolling window w . Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A 21-day volatility estimate responds to a shock faster than a 252-day estimate, but its sample skew and kurtosis can be dominated by one day.

PYTHON / SYSTEM IMPLEMENTATION

Compute strictly backward-looking windows, require minimum observations, retain window endpoints, and visualize multiple defensible windows without choosing after seeing outcomes.

CONFUSIONS TO AVOID

Centered rolling windows leak future data. Overlapping estimates are highly dependent and should not be treated as independent evidence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature pipeline produces point-in-time rolling moments; monitoring compares them with training ranges and triggers review rather than automatic retraining.

RELATED CONCEPTS AND QUICK RECALL

Related: window length, drift, point-in-time feature. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Exponentially weighted moments

BEGINNER

ONE-SENTENCE MEANING

Exponentially weighted estimators assign more weight to recent observations, allowing faster adaptation while retaining a smooth memory of older data.

MEMORY JOGGER

Each step decays old influence by λ ; half-life expresses how quickly weight falls to one half.

WHY IT MATTERS IN QUANT RESEARCH

EW mean and variance are common for volatility forecasting and online monitoring. Their effective sample size is smaller than the raw observation count.

FORMULA INTUITION

w_j proportional to λ^j ; half-life = $\log(0.5)/\log(\lambda)$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

With a 20-day half-life, an observation 20 days old has half the weight of today's observation before normalization.

PYTHON / SYSTEM IMPLEMENTATION

Use one explicit recursion and initialization rule, normalize weights consistently, and log λ or half-life. Compare batch and streaming outputs in tests.

CONFUSIONS TO AVOID

Different packages use adjust, bias, and initialization conventions that materially change early estimates. Recent weighting does not automatically handle structural breaks.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The online risk service updates EW variance with event-time controls and publishes estimator age, effective sample size, and warm-up status.

RELATED CONCEPTS AND QUICK RECALL

Related: EWMA, half-life, effective sample size. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Cross-sectional versus time-series moments

BEGINNER

ONE-SENTENCE MEANING

Time-series moments summarize one entity across dates, while cross-sectional moments summarize many entities at one date. Their populations, dependence, and interpretations differ.

MEMORY JOGGER

Across time asks how an asset behaves; across assets asks how the market is distributed now.

WHY IT MATTERS IN QUANT RESEARCH

Signal normalization, universe breadth, dispersion trades, and risk monitoring require separating these two axes. Mixing them can generate nonsensical features and leakage.

FORMULA INTUITION

time-series: statistic_i over t; cross-section: statistic_t over i. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

The skew of 500 stock returns today describes market breadth asymmetry; the skew of one stock's last 500 daily returns describes its historical payoff shape.

PYTHON / SYSTEM IMPLEMENTATION

Name axes explicitly, preserve as-of membership, and use group-aware operations. Unit-test with small panels where expected axis results differ visibly.

CONFUSIONS TO AVOID

Using current index members for historical cross-sections creates survivorship bias. Flattening the panel blends time and asset dependence into an uninterpretable statistic.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The panel-data layer exposes axis-safe transformations and historical universe membership to both alpha and risk pipelines.

RELATED CONCEPTS AND QUICK RECALL

Related: panel data, universe definition, survivorship. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Missing data, stale prices, and moment bias

BEGINNER

ONE-SENTENCE MEANING

Missingness and stale observations alter moment estimates when they are related to liquidity, volatility, distress, or market closures. Zero-filling is usually an economic assumption, not a neutral cleaning step.

MEMORY JOGGER

Ask why the value is missing before choosing a treatment; the cause may be part of the risk signal.

WHY IT MATTERS IN QUANT RESEARCH

Illiquid assets can show artificially low variance and distorted higher moments when unchanged prices are treated as fresh observations. Delistings can erase negative tails.

FORMULA INTUITION

observed statistic targets $P(X \mid \text{observed})$, which may differ from $P(X)$. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

Forward-filling a weekly quote into daily data creates repeated zero returns followed by a jump, inflating kurtosis and distorting autocorrelation.

PYTHON / SYSTEM IMPLEMENTATION

Track observation age and missing reason, align trading calendars, and compare complete-case, stale-aware, and model-based treatments. Preserve delisting returns.

CONFUSIONS TO AVOID

Mean imputation compresses variance; zero fill can invent outcomes; dropping missing rows changes the sampled population. No method is universally harmless.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The data-quality layer emits freshness and missingness features that gate eligibility before moments enter the feature store or risk model.

RELATED CONCEPTS AND QUICK RECALL

Related: missing-not-at-random, stale quote, delisting bias. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Leakage-safe moment features

BEGINNER

ONE-SENTENCE MEANING

A moment used as a predictive feature must be calculated only from data available before the decision timestamp. Window definition, adjustment data, and universe membership are part of that information boundary.

MEMORY JOGGER

Every rolling mean or volatility is a fitted statistic with a cutoff; future rows must be unable to change past feature values.

WHY IT MATTERS IN QUANT RESEARCH

Moment features are common in signals, scaling, and regime detection. Leakage can create smooth, plausible backtests that are impossible to reproduce live.

FORMULA INTUITION

`feature_t = statistic(x_s: s <= t-lag)`, with lag matching execution timing. Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A close-to-close volatility used to trade at the same close is unavailable unless the execution assumption permits observing and acting on the final print.

PYTHON / SYSTEM IMPLEMENTATION

Use event-time joins, explicit lags, immutable cutoffs, and a perturbation test that changes future data and verifies all past features remain identical.

CONFUSIONS TO AVOID

Full-sample standardization, centered windows, revised data, and today's constituents are frequent leakage routes even when model training itself is walk-forward.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature store enforces availability timestamps and lineage, while validation replays the exact online calculation for each historical decision.

RELATED CONCEPTS AND QUICK RECALL

Related: look-ahead bias, event time, walk-forward validation. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Streaming moments and numerical stability

BEGINNER

ONE-SENTENCE MEANING

Streaming algorithms update moments without retaining every observation. Stable recursions avoid the cancellation and overflow risks of naive sums of powers.

MEMORY JOGGER

Update centered aggregates incrementally; do not subtract enormous raw totals to recover tiny dispersion.

WHY IT MATTERS IN QUANT RESEARCH

Live risk, execution monitoring, and large data pipelines need online mean, variance, skewness, and kurtosis with deterministic merge behavior.

FORMULA INTUITION

$\text{mean}_n = \text{mean}_{(n-1)} + (x_n - \text{mean}_{(n-1)})/n$; use paired recursions for M_2, M_3, M_4 . Check the estimator, units, horizon, and existence assumptions before interpreting it.

SIMPLE MARKET EXAMPLE

A feed with prices near 100,000 and cent-level changes can lose variance precision if computed as $\text{mean}(x^2) - \text{mean}(x)^2$.

PYTHON / SYSTEM IMPLEMENTATION

Use a reviewed Welford/Pebay implementation, merge partial aggregates in tests, handle NaN and overflow explicitly, and compare with high-precision batch results.

CONFUSIONS TO AVOID

A stable recursion does not solve bad timestamps, duplicate events, changing units, or nonstationarity. Reset and warm-up policies need economic definitions.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The streaming statistics service publishes aggregate state, schema version, and replay checkpoints for live monitors and batch reconciliation.

RELATED CONCEPTS AND QUICK RECALL

Related: Welford algorithm, online aggregation, floating point. Explain the statistic in plain English, identify its strongest assumption, and name one diagnostic that would stop you from trusting it.

Same center, different shape

BEGINNER

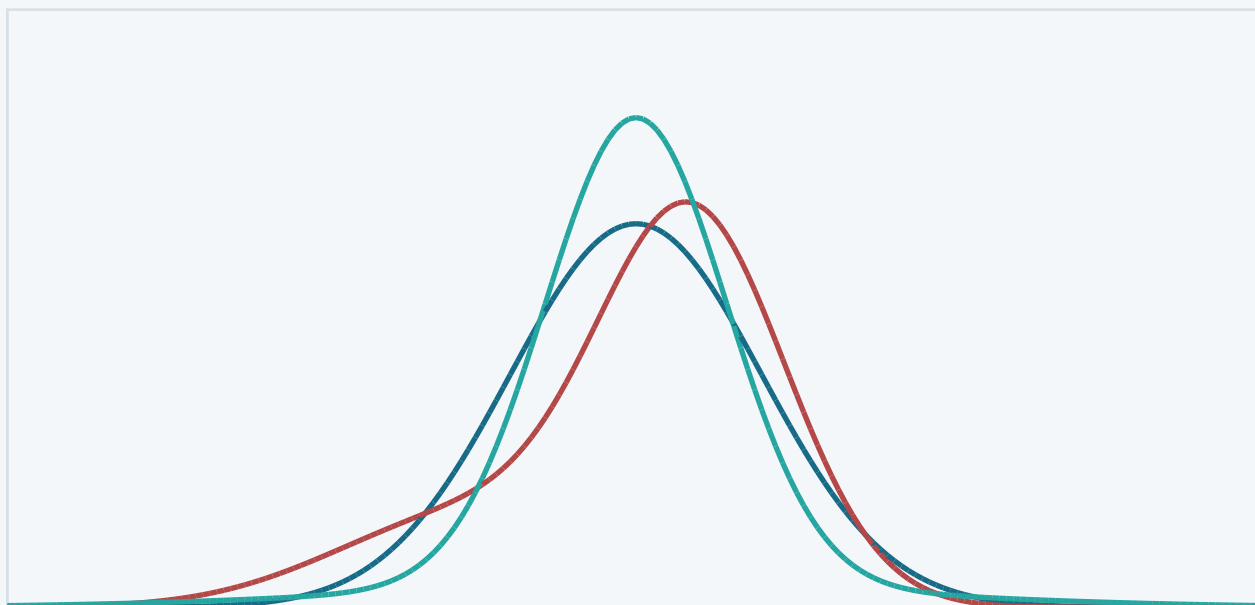
MOMENT SEQUENCE

Mean locates center, variance sets scale, skewness captures directional asymmetry, and kurtosis reacts to extreme standardized deviations. No finite set of moments uniquely shows the full distribution.

Symmetric reference

Left-skew mixture

Heavy-tail mixture



READING THE CURVES

The red mixture puts disproportionate mass in the left tail and tends toward negative skew. The cyan mixture is symmetric but includes a wide stress component, producing high kurtosis without directional skew.

QUANT INTERPRETATION

Two strategies can share mean and variance while differing in crash exposure. Skewness distinguishes direction, but tail quantiles and expected shortfall are still needed to locate economically relevant losses.

DIAGNOSTIC PRACTICE

Overlay an ECDF, mark influential observations, and condition on regimes. Use the moment panel to pose questions, then answer them with distribution and event-level evidence.

RECALL

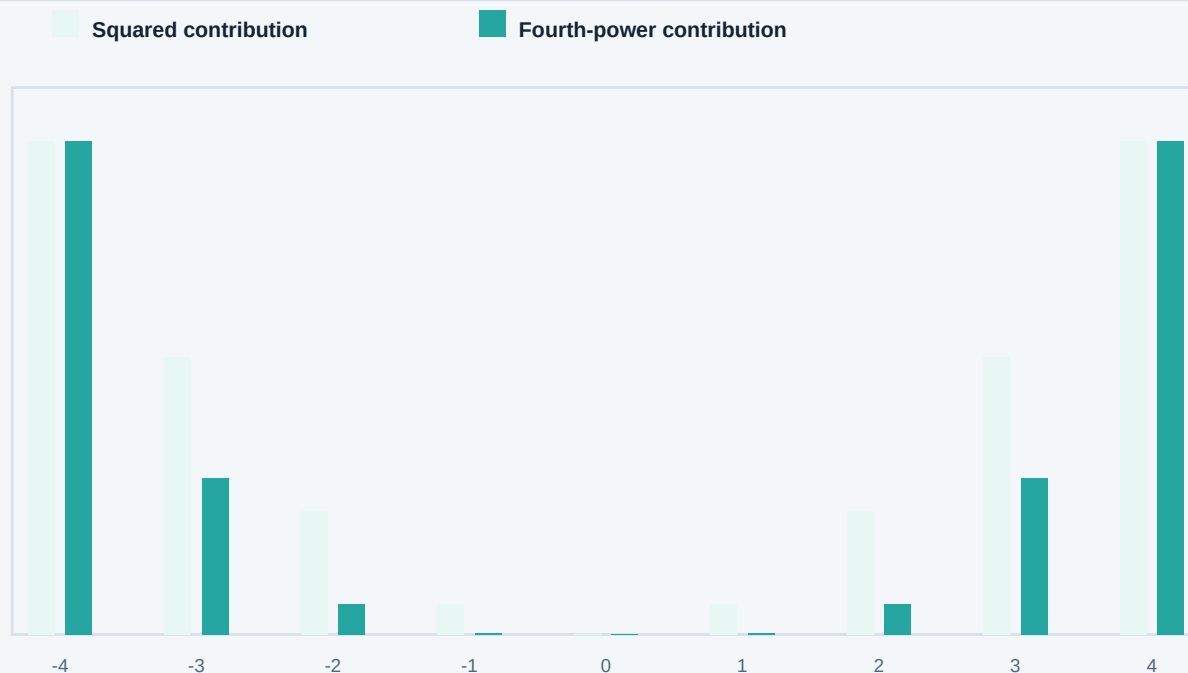
A symmetric distribution may have extreme kurtosis; a high win rate may have negative skew; a positive mean may coexist with both. Each statement concerns a different property.

Power contributions and outlier influence

BEGINNER

WHY HIGHER MOMENTS JUMP

After centering, contribution grows with deviation raised to the moment order. A few distant observations can therefore explain nearly all sample skewness or kurtosis.



WORKED RATIO

Compare deviations 4 and 2. Their variance contributions differ by $4^2/2^2=4$, while their fourth-moment contributions differ by $4^4/2^4=16$. Extreme values dominate increasingly fast at higher order.

INVESTIGATION ORDER

First verify the raw record, adjustment, timestamp, and instrument. Next classify the event as error or genuine. Then report influence and sensitivity; never remove a valid shock merely to stabilize a metric.

SYSTEM CONTROL

Persist per-row contributions and the top contributors with each metric artifact. When source data change, impact analysis can identify exactly which published moments need recomputation.

RECALL

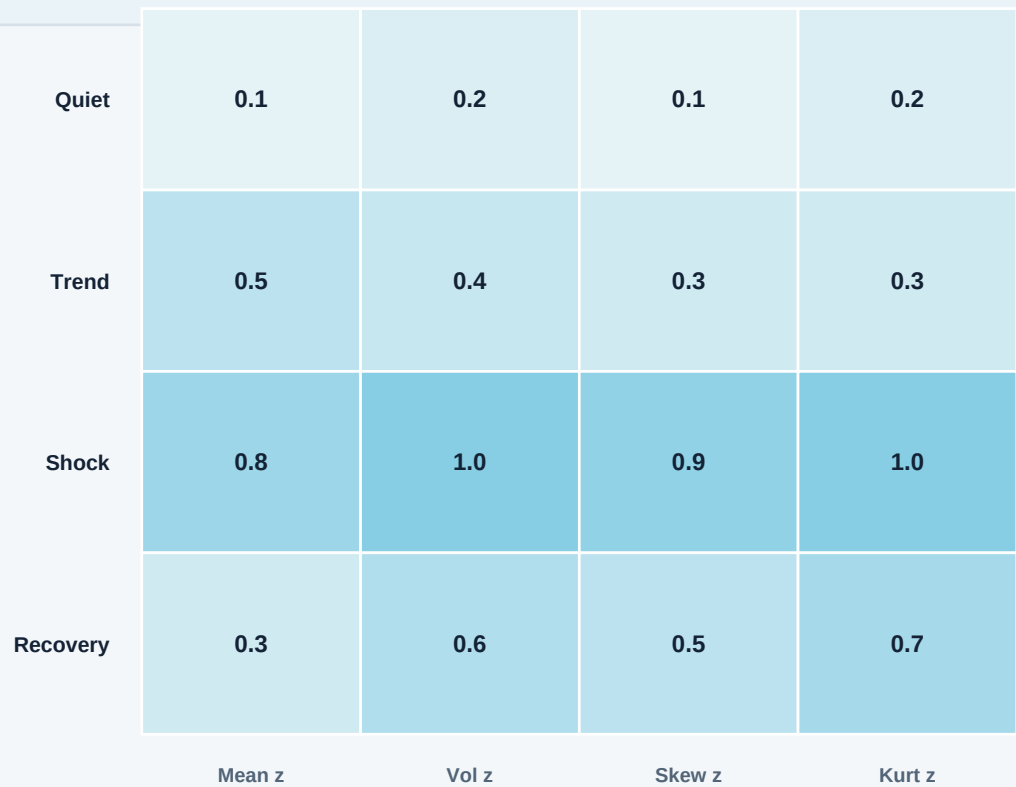
Higher-moment instability can be genuine evidence about tail behavior. The goal is not a smooth number; it is a correct, traceable description of uncertainty.

Moment stability across regimes

BEGINNER

READ BY ROW AND COLUMN

Rows represent market regimes and columns show standardized diagnostics. Color intensity indicates distance from the training reference, so the map reveals which moment is changing and where.



PATTERN INTERPRETATION

Shock periods shift every moment, with volatility and kurtosis furthest from reference. Recovery retains elevated kurtosis after mean and volatility begin normalizing, suggesting lingering event risk or a mixed population.

VALIDATION ACTIONS

Compare raw and robust measures, inspect contribution dates, refit only through approved walk-forward rules, and test downstream performance under stress. A drift alert starts investigation; it does not automatically justify retraining.

GUARDRAIL

Reference ranges must be fit on historical training data. Defining regimes or thresholds with the full sample turns a monitoring visual into a hindsight summary.

Formula sheet and worked calculations

BEGINNER

DISCRETE EXPECTATION

For outcomes $\{-2, 0, 3\}$ with probabilities $\{0.20, 0.50, 0.30\}$, $E[X]=0.5$. The second raw moment is 3.5, so variance is $3.5 - 0.25 = 3.25$ and standard deviation is 1.803.

WEIGHTED EXPECTATION

Forecasts $\{4, 1, -2\}$ bps with probabilities $\{0.25, 0.50, 0.25\}$ give 1.0 bp. Contributions $\{1.0, 0.5, -0.5\}$ show that the positive edge relies on the first state.

PORTFOLIO VARIANCE

For weights w and covariance Σ , variance is $w^T \Sigma w$. Compute Σw first, multiply elementwise by w for component contributions, and verify their sum equals total variance.

SKEWNESS CONTRIBUTION

Center each return, cube it, and divide the aggregate by σ^3 . Preserve the sign: large negative deviations contribute negatively, unlike variance and kurtosis.

KURTOSIS CONVENTION

If standardized fourth moment is 7.4, Pearson kurtosis is 7.4 and excess kurtosis is 4.4. The normal benchmarks are 3 and 0 respectively.

TOTAL VARIANCE

With state weights $\{0.75, 0.25\}$, conditional variances $\{1, 9\}$, and means $\{0.2, -0.6\}$, within-state variance is 3; between-state variance is 0.12; total is 3.12.

Failure modes and diagnostic table

BEGINNER

MEAN RISES, MEDIAN DOES NOT

Inspect rare positive contributors, state weights, data revisions, and selection. Report contribution concentration and confidence rather than claiming broad improvement.

VOLATILITY LOOKS TOO LOW

Check stale prices, zero fills, calendar alignment, smoothing, missing observations, and estimator window. Compare close-to-close with intraday range and scenario risk.

SKEW SIGN FLIPS REPEATEDLY

Measure sample size, top cubic contributors, rolling sensitivity, and regime mix. Treat the estimate as unstable until a mechanism and repeatable evidence emerge.

KURTOSIS EXPLODES

Verify corporate actions and bad ticks, then inspect genuine shock dates and tail index. Do not winsorize automatically; separate data correction from risk transformation.

ANNUALIZED RISK MISSES REALIZED RISK

Test autocorrelation, volatility clustering, overlapping returns, varying calendars, and compounding. Prefer direct horizon estimates or a temporal model.

LIBRARIES DISAGREE

Compare divisor, bias correction, NaN handling, weights, window endpoints, and Pearson versus excess kurtosis. Pin definitions in the metric catalog.

Implementation and metric-registry checklist

BEGINNER

VARIABLE CONTRACT

Outcome name, unit, horizon, event clock, population, eligibility, return convention, adjustment policy, missing treatment, and owner.

ESTIMATOR CONTRACT

Population or sample, divisor, weights, half-life, window, minimum n, bias correction, interpolation, robust transform, and supported moment orders.

POINT-IN-TIME CONTROLS

Availability timestamp, lag, historical universe, revision policy, immutable source hash, train cutoff, and future-perturbation leakage test.

NUMERICAL CONTROLS

Stable centered algorithm, overflow and NaN behavior, merge tests, decimal-versus-percent tests, analytic toy cases, and high-precision comparison.

VALIDATION EVIDENCE

Effective sample size, bootstrap range, rolling stability, regime panels, top contributors, tail diagnostics, robust challenger, and downstream sensitivity.

RUNTIME GOVERNANCE

Metric ID, code and library versions, freshness, warm-up status, drift thresholds, alert playbook, fallback, approval, and rollback artifact.

Final active recall index

BEGINNER

EXPECTATION

Define discrete and continuous expectation, linearity, conditional expectation, tower property, Jensen's inequality, net edge, and the existence condition.

VARIANCE

Explain squared deviation, computational identity, sample divisor, scaling, covariance terms, portfolio quadratic form, total variance, and downside variance.

SKEWNESS

State the standardized third-moment formula, interpret sign without referring to win rate, identify influential observations, and explain estimator instability.

KURTOSIS

Contrast Pearson and excess conventions, reject the simple peakedness myth, connect mixture regimes to high kurtosis, and check fourth-moment existence.

ESTIMATION PRACTICE

Compare rolling, expanding, exponentially weighted, robust, cross-sectional, time-series, batch, and streaming calculations with their failure modes.

PROMOTION TEST

Before publishing a moment, answer: what is X , when was it knowable, does the moment exist, which estimator was used, who drives it, how stable is it, and what decision consumes it?